

Research and Teaching Statement

Paul Goldsmith-Pinkham · September 2026

Who benefits when lenders become better at predicting default? Does debt relief help households recover from financial distress? Why do households participating in the same housing market experience such different outcomes? How can we credibly provide causal answers to these and other economic questions?

My research tackles this agenda by contributing to household finance and applied econometrics literatures. Often, my empirical work motivates research in applied econometrics – I think of myself as a household finance researcher and applied econometrician, emphasis on *applied*. For example, in my work on consumer bankruptcy, we used differences in judges' willingness to grant protection to estimate the effects of debt relief. Working with this design also helped spark my research on multiple treatment designs and how judge-based instruments should be constructed and interpreted. Similarly, my methodological interests have expanded the set of household finance problems I study, such as using my experience with econometric modeling to work on the effects of machine learning on credit markets.

I have published 18 peer-reviewed papers since my PhD (six on Covid-related topics), 16 of them since arriving at Yale SOM in 2018, with four more conditionally accepted or at the revise-and-resubmit stage. As of September 2026, I was ranked by RePec in the top 100 authors based on publications in the last ten years. My research has reached beyond academia: it has been covered by The Economist, NPR, CNBC, and Forbes, and several papers are assigned in PhD (*Bartik Instruments*, *Predictably Unequal*, *Macroeconomic Effects of Debt Relief*) and MBA courses (*Predictably Unequal*) at other universities.

Household Finance

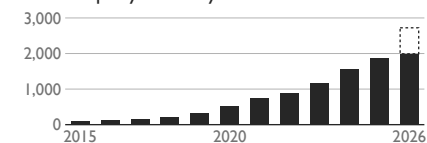
The effect of information and technology

My first line of research studies the role of information and technology in consumer lending. In *Predictably Unequal? The Effects of Machine Learning on Credit Markets*,¹ we study the distributional consequences of the adoption of improved machine learning techniques in the domain of mortgage markets. Lenders have increasingly used these techniques in different credit settings with the goal of improving predictive accuracy.

The idea underlying our paper is that a more sophisticated statistical technology produces predictions with greater variance than a more primitive technology. This means that there will always be some borrowers considered less risky by a new credit scoring technology, or “winners”, while other borrowers will be deemed riskier (“losers”), relative to their position under the pre-existing technology. Using data on US mortgages, we predict default using traditional and machine learning models. We find that Black and Hispanic borrowers are disproportionately less likely to gain from the introduction of machine learning.

In the United States, credit reports — and the associated credit scores — are used in nearly all consumer lending decisions. They have increasingly

Citations per year to my work



Google Scholar, since 2015; 10,153 citations in total as of 24 September 2026. In this and the charts below, the 2026 bar is solid through 24 September and dashed to its annualized value.

¹ Predictably Unequal? The Effects of Machine Learning on Credit Markets
Andreas Fuster, Paul Goldsmith-Pinkham, Tarun Ramadorai, and Ansgar Walther
Journal of Finance 77(1) (2022), 5–47
[doi:10.1111/jofi.13090](https://doi.org/10.1111/jofi.13090)

★ Brattle Group Prize in Corporate Finance, First Place, 2022

★ WFA Wharton-WRDS Best Empirical Finance Paper, 2019

Selected press mentions

• The Economist, February 2019



Citations per year, 2018–2026; 1,364 in total.

been used in a variety of other settings such as job applications. In *Bad Credit, No Problem?*², we study consumer bankruptcy flags on credit reports, which is one of the consequences of filing for bankruptcy in the United States. We show that the removal of these negative “flags” on consumer credit reports increased individuals’ access to credit but had no impact on their employment or income. This research can inform the consequences of policies limiting the use of credit reports in employment decisions and contributes to our understanding of the costs and benefits of the bankruptcy system.

Financial protection and social insurance

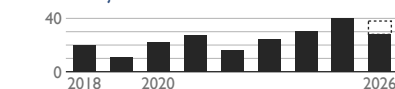
Much of my research focuses on understanding government programs, such as consumer bankruptcy and Medicare, which provide a form of financial protection.

To evaluate the efficiency of the bankruptcy system, one crucial piece is to quantify the “ex post” *individual* benefits of the bankruptcy system. In *Consumer Bankruptcy and Financial Health*,³ we show that the debt forgiveness from a successful bankruptcy discharge lowers the probability of adverse post-bankruptcy financial events and increases the probability of subsequent home ownership. Identifying the causal channel is challenging, since the individuals who chose to file are very different than non-filers. We use the random assignment of bankruptcy judges, who differ in their leniency to discharge cases, to instrument for successful bankruptcy discharge and quantify these effects.

*Macroeconomics Effects of Debt Relief*⁴ connects household relief to *aggregate* outcomes. In the paper, we combine empirical estimates with a structural macroeconomic model to guide the interpretation of the estimated effects. First, we show empirically that states with more generous bankruptcy exemptions had significantly smaller declines in non-tradable employment and larger increases in unsecured debt write-downs compared to states with less generous exemptions. We interpret these reduced form estimates as *relative* debt relief effects across states. However, to get the overall *aggregate* macroeconomic effects, we need to deal with an econometric issue known as the “missing intercept” problem. Crucially, the overall aggregate effect of debt relief can only be identified with additional assumptions on the interactions across states and sectors. Using the model, we show that the debt relief induced by bankruptcy protections lead to higher aggregate employment in *both* tradable and non-tradable employment, despite no relative different in tradable employment, since debt relief stimulus has effects on tradable employment *across* regions.

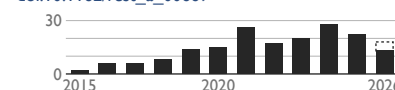
With Maxim Pinkovskiy and Jacob Wallace, I study health insurance as another form of financial protection.⁵ Medicare eligibility at age 65 reduces debt in collections by roughly 30 percent and geographic variation in collections by about two-thirds. A crucial non-result in our paper is that we find almost no effects of Medicare on reducing other adverse financial conditions — credit scores, delinquent debts, bankruptcy, etc. — beyond debts in collection. This fits into a large policy debate on the purpose and role of health insurance, suggesting that at least among 65-year-olds, the benefits of expanded Medicare to financial health are quite limited. Our approach

- ² **Bad Credit, No Problem? Credit and Labor Market Consequences of Bad Credit Reports**
Will Dobbie, Paul Goldsmith-Pinkham, Neale Mahoney, and Jae Song
Journal of Finance 75(5) (2020), 2377–2419
doi:10.1111/jofi.12954



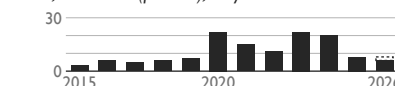
Citations per year, 2018–2026; 231 in total.

- ³ **Consumer Bankruptcy and Financial Health**
Will Dobbie, Paul Goldsmith-Pinkham, and Crystal S. Yang
Review of Economics and Statistics 99(5) (2017), 853–869
doi:10.1162/rest_a_00669



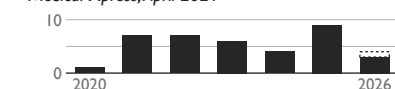
Citations per year, 2015–2026; 178 in total.

- ⁴ **Macroeconomic Effects of Debt Relief: Consumer Bankruptcy in the Great Recession**
Adrien Auclert, Will Dobbie, and Paul Goldsmith-Pinkham
American Economic Review. *Revised and resubmitted*. NBER Working Paper 25685.
Selected press mentions
• Vox, *The Weeds* (podcast), May 2019



Citations per year, 2015–2026; 133 in total.

- ⁵ **The Great Equalizer: Medicare and the Geography of Consumer Financial Strain**
Paul Goldsmith-Pinkham, Maxim Pinkovskiy, and Jacob Wallace
Review of Economics and Statistics. *Conditionally accepted*, March 2026.
Selected press mentions
• *Medical Xpress*, April 2021



Citations per year, 2020–2026; 37 in total.

uses a regression discontinuity approach at age 65 combined with a rich microdata set to estimate the treatment effect of Medicare across more than 700 commuting zones, which allows us to identify which areas benefit the most, and how the benefits of Medicare have changed due to the roll-out of the Affordable Care Act.

Housing returns and intermediation

My housing research asks why households participating in the same market can accumulate different amounts of wealth. In *The Gender Gap in Housing Returns*,⁶ we investigate the gender gap in financial returns from housing investments in the United States between 1991 and 2017. Using a large dataset of over 50 million housing transactions, we find that single men earn 1.5 percentage points higher unlevered annualized returns compared to single women. About 45% of this gap is attributed to market timing, i.e., the strategic choice of where and when to buy and sell properties.

The remaining gap is largely driven by differences in execution prices at the points of purchase and sale, rather than investment in upgrades or riskier property choices. The gap narrows in tighter markets, where traditional bilateral negotiations give way to quasi-auctions. This suggests that negotiation dynamics play a significant role in perpetuating the gender gap in housing investment returns.

We focus on the role of the real estate agent as an intermediary affecting housing transactions in *Heterogeneous Real Estate Agents and the Housing Cycle*.⁷ Using data on 8.5 million listings in the United States from 2001 to 2014, we show that listings of inexperienced real estate agents are less likely to sell than those of experienced agents. We estimate this using a novel instrumental variables identification strategy that exploits the stickiness of real estate agents — people are likely to use the agent they bought with to subsequently sell their home — and the exit of real estate agents. We then use a dynamic search model to show that the ubiquity of these inexperienced agents has aggregate implications for the average sale probability of listed properties.

Applied Econometrics

Identifying and interpreting economic effects

My research in econometrics caters to the applied researcher. I have written papers on the identification assumptions behind commonly used empirical methods, and have developed new methods to address common empirical challenges.

*Bartik Instruments: What, When, Why, and How*⁸ came from an effort to understand the identification conditions behind a Bartik'' or shift-share instrument, a popular research approach in finance and numerous other fields. A fundamental question that long eluded attention is: when is a Bartik instrument valid? Concretely, in the China trade shock, is the identification assumption about the growth rate of imports (the shift), or about the industry composition of locations (shares)? In our paper, we open the black box of the Bartik instrument by formalizing its structure and providing tools to clarify the approach into more traditional comparisons.

⁶ The Gender Gap in Housing Returns

Paul Goldsmith-Pinkham and Kelly Shue

Journal of Finance 78(2) (2023), 1097–1145

doi:10.1111/jofi.13212

★ WFA Wharton-WRDS Best Empirical Finance Paper, 2020

Selected press mentions

• NPR Planet Money (*The Indicator*), January 2020

• CNBC, January 2020

• The Economist, February 2020



Citations per year, 2020–2026; 198 in total.

⁷ Heterogeneous Real Estate Agents and the Housing Cycle

Sonia Gilbukh and Paul Goldsmith-Pinkham

Review of Financial Studies 37(11) (2024), 3431–3489

doi:10.1093/rfs/hhae048

Selected press mentions

• NPR Planet Money (*The Indicator*)

• Forbes, May 2021

• Freakonomics Radio, January 2025



Citations per year, 2017–2026; 96 in total.

⁸ Bartik Instruments: What, When, Why, and How

Paul Goldsmith-Pinkham, Isaac Sorkin, and Henry Swift

American Economic Review 110(8) (2020), 2586–2624

doi:10.1257/aer.2018.1047



Citations per year, 2017–2026; 4,252 in total.

We show that the usual Bartik estimator can be represented as a weighted combination of the underlying exposure-based instrumental-variables estimates. The associated Rotemberg weights reveal which industries or exposures drive the result. Under the identification framework we study, researchers can assess whether those particular shares are plausibly unrelated to omitted determinants of the outcome. The weights direct attention to the comparisons that matter for the estimate, rather than treating every component of the instrument as equally informative.

In *Contamination Bias in Linear Regressions*,⁹ we study linear regressions, a workhorse estimation tool in empirical economics. We show when researchers are interested in estimating the effect of multiple treatments X on outcomes Y (e.g., the effect of multiple randomized treatments, or even observational differences across groups) after controlling for a set of covariates W , the standard approach of directly including the controls, and then additively estimating the treatment effects, can be subject to bias when there are heterogeneous effects (e.g. when estimating $Y = \beta X + \gamma W + \varepsilon$, the coefficient β can be biased). This estimation approach shows up in many common empirical examples such as randomized experiments, value-added models, difference-in-differences and judge leniency designs. We then provide simple corrections that can be applied in practice to account for this bias.

In related work, *Leniency Designs: An Operator's Manual*,¹⁰ we address implementation and interpretation in studies using assignment to judges and other decision-makers. My bankruptcy research helped motivate this work. We provide a jackknife estimator, diagnostics, and guidance for inference. In the paper, we replicate an existing paper in the *Journal of Finance*, “*What Is a Patent Worth? Evidence from the U.S. Patent “Lottery”*”, a key paper in the patent and innovation literature and demonstrate the usefulness of the tools in this setting.

Spillovers and networks

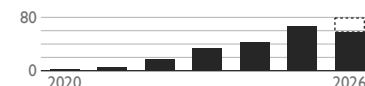
I additionally do methodological research on spillovers and networks. With Guido Imbens,¹¹ we show that canonical peer effect models need to account for the selection **into** peer groups. With Arun Chandrasekhar, Tyler McCormick, Samuel Thau, and Jerry Wei, we study *diffusion* processes on networks (e.g. contagion like COVID-19, or information processes like in a bank run), and show that when an econometrician has a tiny fraction of missing network links, this can produce large diffusion forecast errors even when aggregate diffusion parameters remain consistently estimable.¹² This is a follow-up on work where we show that regional authorities are unable to coordinate lockdowns during a pandemic like COVID-19 due to spillovers across regions and delays in detection.¹³

Selected work in progress

I have a number of ongoing papers touching on household finance and applied econometrics. I outline two of them below.

My work with Tianshu Lyu (a former Yale SOM PhD student) on Causal Inference in Financial Event Studies¹⁴ extends my econometric agenda to a central empirical procedure in finance. Financial event studies compare a

- ⁹ Contamination Bias in Linear Regressions
Paul Goldsmith-Pinkham, Peter Hull, and Michal Kolesár
American Economic Review 114(12) (2024), 4015–4051
doi:10.1257/aer.2022.1116



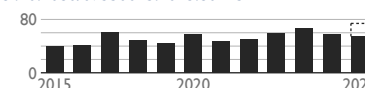
Citations per year, 2020–2026; 228 in total.

- ¹⁰ Leniency Designs: An Operator's Manual
Paul Goldsmith-Pinkham, Peter Hull, and Michal Kolesár
Journal of Economic Perspectives 40(3) (2026), 213–240
doi:10.1257/jep.2025.1480



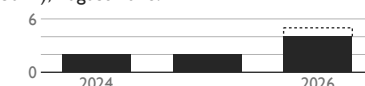
Citations per year, 2025–2026; 17 in total.

- ¹¹ Social Networks and the Identification of Peer Effects
Paul Goldsmith-Pinkham and Guido W. Imbens
Journal of Business & Economic Statistics 31(3) (2013), 253–264
doi:10.1080/07350015.2013.801251



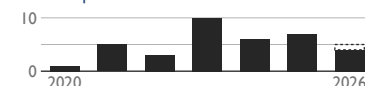
Citations per year, since 2015; 683 in total.

- ¹² Non-robustness of Diffusion Estimates on Networks with Measurement Error
Arun Chandrasekhar, Paul Goldsmith-Pinkham, Tyler McCormick, Samuel Thau, and Jerry Wei
Econometrica. Revise and resubmit (second round), August 2026.



Citations per year, 2024–2026; 8 in total.

- ¹³ Interacting Regional Policies in Containing a Disease
Arun G. Chandrasekhar, Paul Goldsmith-Pinkham, Matthew O. Jackson, and Samuel Thau
Proceedings of the National Academy of Sciences 118(19) (2021), e2021520118
doi:10.1073/pnas.2021520118



Citations per year, 2020–2026; 36 in total.

- ¹⁴ Causal Inference in Financial Event Studies
Paul Goldsmith-Pinkham and Tianshu Lyu
Working paper, August 2026.



Citations per year, 2025–2026; 9 in total.

security's return around an event with a benchmark return, often an asset pricing model like the CAPM or Fama-French 3 Factor model. We find that the standard approach in finance will be biased if the benchmark portfolio, like the market index, omits a common source of return variation. The exception is when there are many events and these events are randomly distributed *in time*. We show a number of new formal results about bias in these settings, and then also propose two alternative estimators drawn from the causal inference literature.

Using data from auto lending asset-backed securities, I am doing work with Fuster, Ramadorai and Walther to use discontinuities in interest rate schedules *across* lenders to study how competition across lenders is affected by differences in pricing schedules, and whether these differences reflect differences in payment prediction models. This work is motivated by the fact that *Predictably Unequal?* was followed by a number of papers by other authors studying the role of both new data and technology on credit lending behavior. However, an outstanding question that remains is how lenders are able to compete within the same market using different models, if these predictions lead to different lending decisions.

Teaching

I have taught an MBA class called Investment Management throughout my time at Yale SOM. Over the time period and different sections, the average evaluation of the course is 4.1 out of 5.0 (department average is 4.0). In the last three years, the average rating is 4.4 (department average is 4.2).

The primary goal of the course is to provide a broad overview of investment management, focusing on the application of finance theory to the issues faced by portfolio managers and investors in general. The goal of the class is to provide tools to give a disciplined way of thinking about investment decisions, while simultaneously framing how we think about the historical performance of markets. Much of the class is focused on equity markets but we also examine fixed income, derivative securities, and alternative asset classes like hedge funds and private equity.

The course provides a mix of formal theoretical tools — revisiting and expanding on the CAPM factor model, and learning other important portfolio models — and facts about the broader structure of the market. The key lesson I try to focus on in the class is the fundamental trade-off between risk and return in well-diversified portfolios; we capstone this discussion by studying the Yale endowment case, which provides a useful setting for students to see how Yale's unique characteristics allow it to bear certain types of risk (and thus be compensated by the market).

I also teach a second-year PhD course in Applied Empirical Methods. I created the course from scratch in Spring 2021, covering a variety of empirical methods used in economics and finance research and focused on causal inference. Across its six offerings, the course has received an average evaluation of 4.25 out of 5.0, with a department average of 4.15. My materials are [posted on Github](#) with significant traffic (over 200 unique visitors per week) and [my lectures on YouTube](#) have 3,580 subscribers (the first lecture has 23 thousand views). This course is quite popular across Yale University,

Investment Management (MGT 544). MBA elective, taught every spring 2019–2026; two sections in 2019 and 2020.

All years, 2019–2026



Last three years, 2024–2026



Overall course rating, 1–5 scale. The notch marks the school-wide average for the same terms.

Applied Empirical Methods (MGMT 737 / ECON 574; ECON 5574 / MGMT 7202 from 2026). Second-year PhD course, taught every spring 2021–2026.

All years, 2021–2026



Overall course rating, 1–5 scale. The notch marks the school-wide average for the same terms.

with many Ph.D. students from the School of Public Health, School of Environment, Political Science department and Economics department enrolled, along with our School of Management students. In total over six years, I have 170 enrolled students.

The philosophy behind this course is to teach technical methods that are focused on intuition, rather than jargon. Often, many econometrics courses are challenging for practitioners to find much intuition, and so empirical researchers fail to participate in the cutting edge of the research frontier. My goal with the class is to help students understand 1) why there are difficult econometric problems in much of our research; 2) how to solve these issues; 3) and most importantly, how to explain these solutions clearly.